

Lesson 11

Function Notation, Domain & Range

f

Functions · 60-minute lesson

Syllabus SL 2.2 · Property of Lynda Badmus Education

Do Now — recall

ON YOUR WHITEBOARD

1. If $f(x) = 2x + 1$, find $f(3)$.
2. If $g(x) = x^2$, find $g(-4)$.
3. What values of x can you NOT put into $1/x$?

Answers (click to reveal)

1. $f(3) = 7$
2. $g(-4) = 16$
3. $x = 0$ (division by zero)

What we're learning today

LEARNING OBJECTIVES

- 1 Use**
function notation and evaluate $f(x)$.
- 2 Define**
the domain (allowed inputs) of a function.
- 3 Define**
the range (possible outputs) of a function.
- 4 Identify**
values that must be excluded from a domain.

SUCCESS CRITERIA — I CAN...

- ✓
substitute a value into $f(x)$ correctly
- ✓
state the domain of a function
- ✓
state the range of a function
- ✓
exclude values that make a function undefined

Notation, domain & range

THE LANGUAGE

$f(x)$ means the output for input x

Domain

the set of allowed inputs (x-values)

Range

the set of possible outputs (y-values)

Evaluate

replace every x with the given value

PICTURE IT

Domain

read left-right (the x-axis)

Range

read bottom-top (the y-axis)

Map

each input gives exactly one output.

Evaluate a function

QUESTION

Given $f(x) = x^2 - 4x$,

find $f(5)$.

SOLUTION

Step 1 replace every x with 5:

$$f(5) = 5^2 - 4(5)$$

Step 2 evaluate:

$$f(5) = 25 - 20 = 5$$

Tip:

Use brackets when you substitute — it prevents sign and order mistakes.

Your turn — evaluate

ANSWER IN YOUR BOOK

1. $f(x) = 3x - 2$, find $f(4)$.
2. $g(x) = x^2 + 1$, find $g(-3)$.
3. $h(x) = 10/x$, find $h(2)$.

Answers (click to reveal)

1. 10
2. 10
3. 5

Finding the domain

WHAT TO EXCLUDE

no $\frac{1}{0}$, no $\sqrt{\text{negative}}$

Fractions

denominator $\neq 0$

Roots

what's under a $\sqrt{}$ must be ≥ 0

Otherwise

domain is all real numbers, $x \in \mathbb{R}$

EXAMPLES

$$1/(x-3)$$

$$x \neq 3$$

$$\sqrt{(x-2)}$$

$$x \geq 2$$

$$x^2 + 1$$

all real x

Find a domain & range

QUESTION

State the domain and range of

$$f(x) = \sqrt{x - 2}.$$

SOLUTION

Domain: the inside of the root must be ≥ 0 :

$$x - 2 \geq 0 \Rightarrow x \geq 2$$

Range: a square root gives values ≥ 0 :

$$f(x) \geq 0$$

So domain $x \geq 2$, range $f(x) \geq 0$.

Exam-style question

The function f is given by $f(x) = 1/(x - 4)$.

- (a) Write down the value of x that must be excluded from the domain. [1]
- (b) State the domain of f . [2]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) $x = 4$ (A1)
- (b) $x \in \mathbb{R}, x \neq 4$ (A1)(A1)

Common mistakes

 **$f(x)$ is not $f \times x$**

$f(x)$ means 'the function applied to x ', not multiplication.

**Brackets when substituting**

$f(-2)$ with x^2 is $(-2)^2 = 4$, not -4 .

**Domain vs range**

Domain = inputs (x); range = outputs (y).

**Excluded values**

Always check for division by zero and roots of negatives.

Mixed practice

QUESTIONS

1. $f(x) = 2x^2 - 1$, find $f(3)$.
2. State the domain of $1/(x + 5)$.
3. State the domain of $\sqrt{x - 7}$.
4. $g(x) = 5 - x$, find $g(-2)$.

Answers (click to reveal)

1. 17
2. $x \neq -5$
3. $x \geq 7$
4. 7

Plenary & exit ticket

KEY FORMULAS

$f(x)$ = output for input x

exclude: $\frac{1}{0}$, $\sqrt{\text{negative}}$

EXIT TICKET — before you leave

1. $f(x) = 4x + 1$, find $f(2)$.

2. Domain of $1/(x - 1)$?

3. Domain of $\sqrt{x + 3}$?

Answers (click to reveal)

1. 9

2. $x \neq 1$

3. $x \geq -3$

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Function notation, domain & range.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 12

Composite & inverse functions — combining functions and reversing them.

$$(f \circ g)(x) = f(g(x))$$